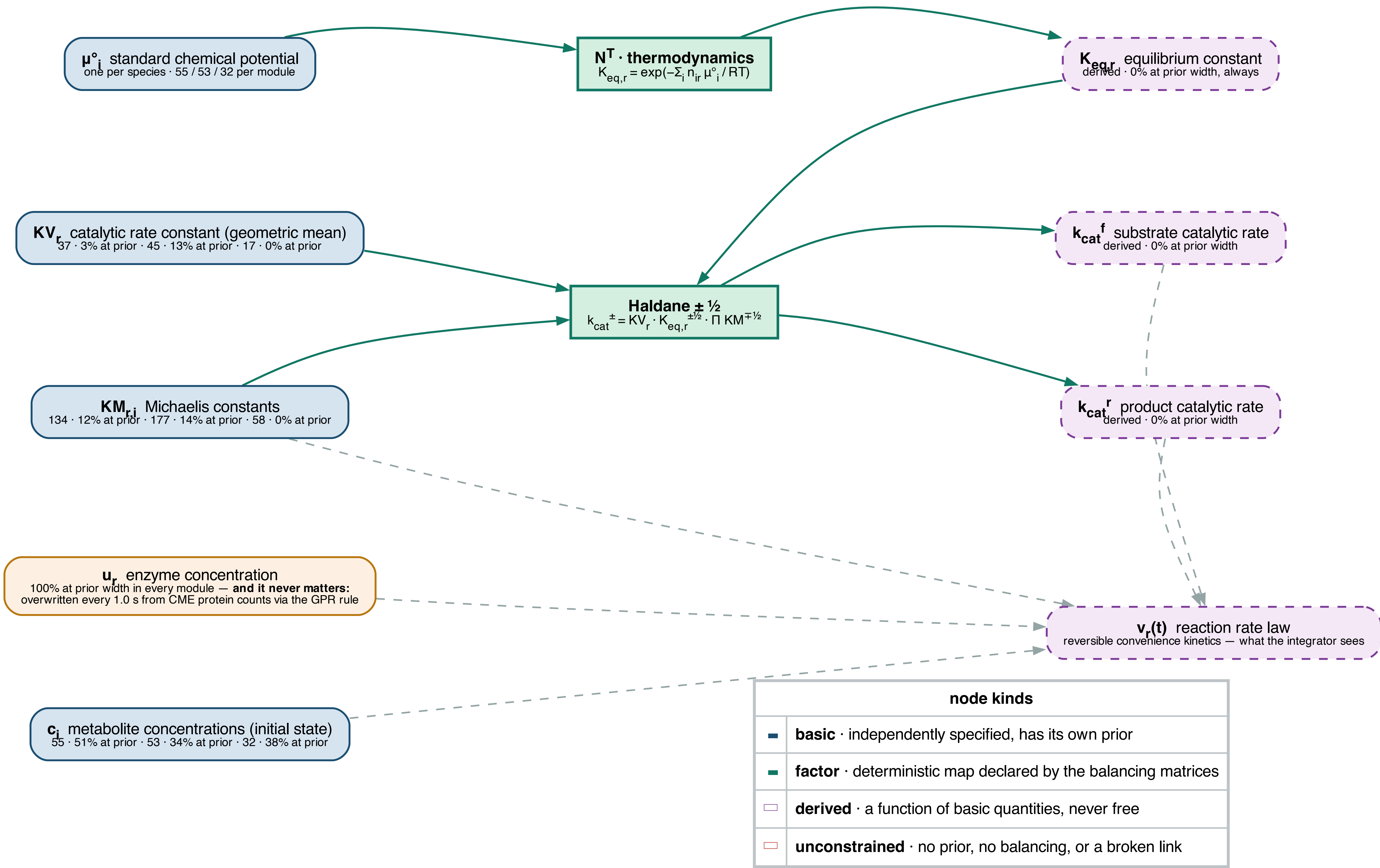


Parameter coupling in the well-stirred minimal cell

Which parameters constrain which — a different graph from the state coupling, over the same modules.
Counts computed over the reactions actually active in the model, not the whole reconstruction.

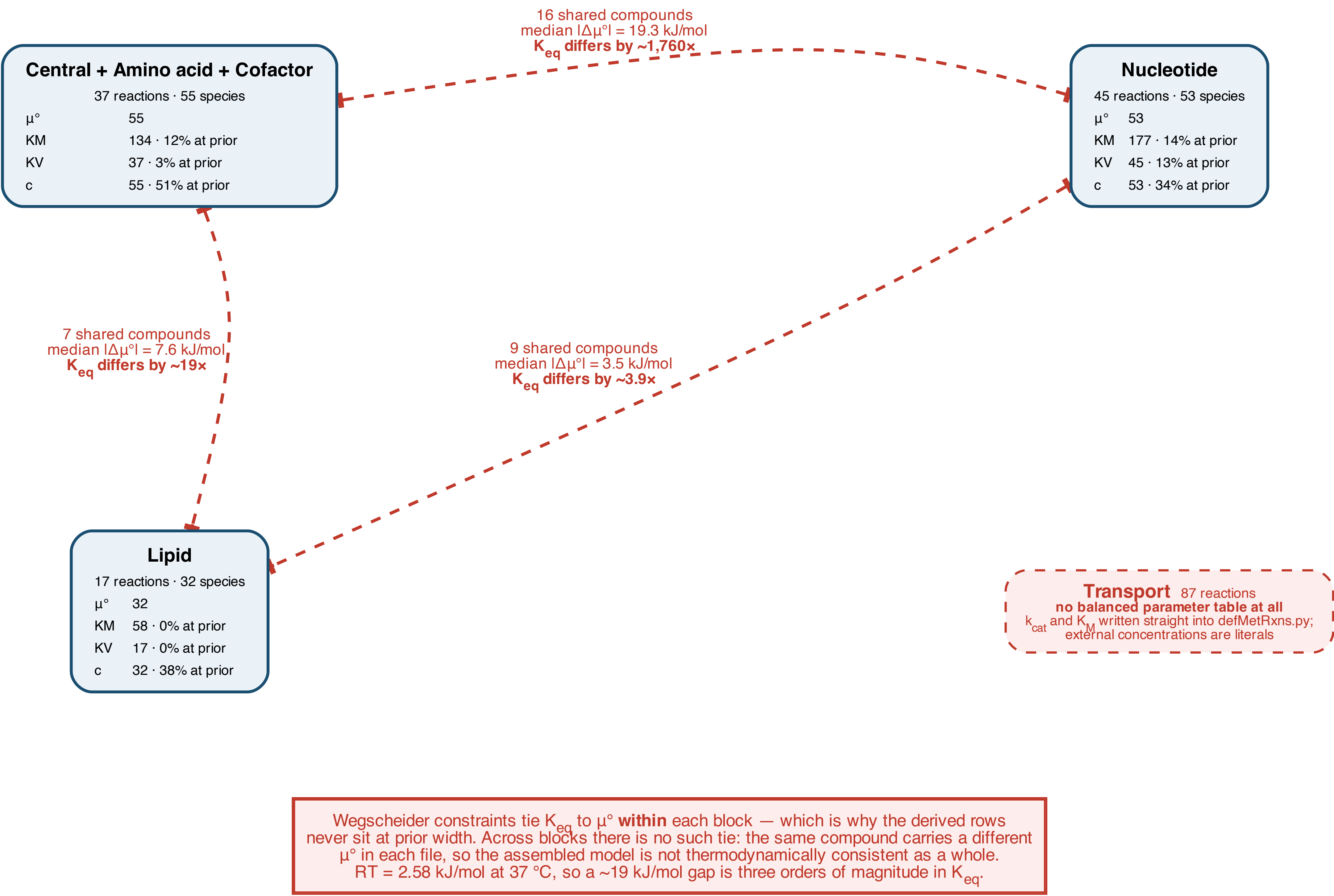
A · Inside one balanced module

the dependency map is declared in each SBtab file's *Parameter prior* MatrixInfo column



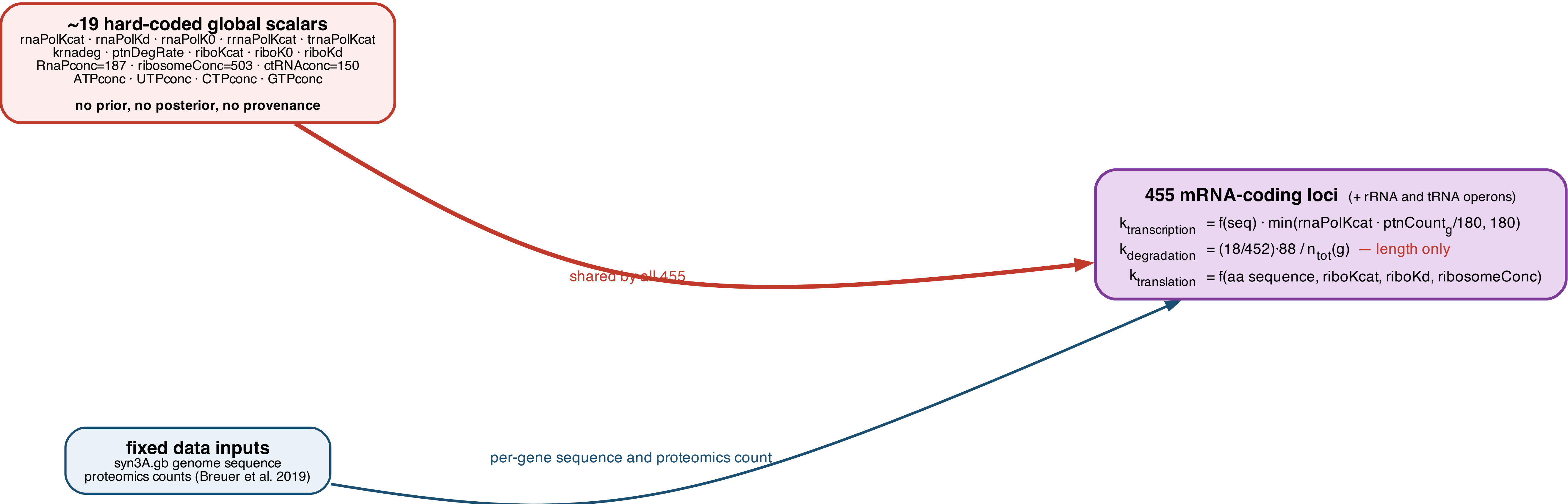
B · Between modules: the constraint that isn't there

each file was balanced separately, so μ° for a shared compound was estimated more than once



C · Gene expression: no factor structure at all

metabolism at least went through balancing; this block did not



Every one of the 455 rate constants is a **deterministic** function of sequence and a handful of shared scalars. The graph is a star, not a chain: perturb one scalar and all 455 rates move together — so the parameters are maximally coupled and individually unidentifiable by construction.

riboKcat is defined twice (10 in MinCell_CMEODE.py, 12 in translation_rate_start.py); the imported module wins, so 10 is dead code.